

EDUCATION

Ph.D. in Physics, Harvard University	Ph.D. Thesis: Hidden Dynamics of Static Friction	2020
M.A. in Physics, Harvard University	Ph.D. Advisor: Shmuel M Rubinstein	2016
B.A. in Physics, Cornell University	Undergraduate Research Advisor: Itai Cohen	2012

RESEARCH EXPERIENCE

ARIA R&D Creator (Independent Researcher)		2025-2027
Topics: Learning Metamaterials Distributed & Analog Machine Learning		
Postdoctoral Fellow University of Pennsylvania	with Douglas J Durian & Andrea J Liu	2020-2025
Topics: Emergent Physical Learning Granular Flows ML in Experimental Science		
Visiting Researcher EPFL, Switzerland	with John M Kolinski	2018
Topic: Developed ultrafast ($\geq$ MHz) imaging technique for any camera		

FUNDING and AWARDS

Grants & Fellowships

ARIA Opportunity Seed Grant	£500k (~\$650k) 2025-2027
AI Fellow, U Pennsylvania	\$8k/year, 2025
Center for Soft and Living Matter Fellow, U Pennsylvania	\$5k/year, 2025
Data Science Postdoctoral Fellow, U Pennsylvania	\$5k/year, 2022-2024
Smith Family Fellowship, Harvard U	\$90k, 2015-16
Purcell Fellowship, Harvard U	\$90k, 2014-15

Research & Teaching Recognition

1 st place GSNP Postdoctoral Presentation Awards	APS March Meeting 2024
1 st place Meeting-Wide Postdoctoral Poster Competition	APS March Meeting 2023
Herbert B. Callen Memorial Prize	University of Pennsylvania 2023
2 nd place, title: Learning Networks on the Radio	MRSEC National Science Slam 2022
Editor's Suggestion, Physical Review Applied	Dillavou et. al. 2022
Rising Star in Soft and Biological Matter	University of Chicago 2021
Physical Review Letters Editor's Suggestion	Dillavou & Rubinstein 2018
Bok Center Certificate of Teaching Excellence	Harvard University Spring 2018

PUBLICATIONS

‡Equal Contribution †Worked performed as an undergraduate

Lead Author

- [1] **S Dillavou**, JW Rocks, JF Wycoff†, AJ Liu, DJ Durian. *Analog Physical Systems Can Exhibit Double Descent.* [arXiv 2511.17825](#)
- [2] **S Dillavou**, S Mishra. *Physical Reinforcement Learning* [arXiv 2511.17789](#)
- [3] **S Dillavou**‡, M Guzman‡, AJ Liu, DJ Durian. *Understanding and Embracing Imperfection in Physical Learning Networks* (in Revision, Science Advances) [arXiv 2505.22887](#)
- [4] JM Hanlan‡, **S Dillavou**‡, AJ Liu, DJ Durian. *Cornerstones are the Key Stones: Using Interpretable Machine Learning to Probe the Clogging Process in 2D Granular Hoppers* [Soft Matter, 2025](#)
- [5] D Hathcock‡, **S Dillavou**‡, JM Hanlan, DJ Durian, Y Tu. *Stochastic dynamics of granular hopper*

- flows: a configurational mode controls the stability of clogs* [Physical Review E, 2025](#)
- [6] (Perspective) **S Dillavou**, *Harnessing Machine Learning to Guide Scientific Understanding* [Physics Magazine, 2024](#)
- [7] **S Dillavou**, B Beyer[†], M Stern, AJ Liu, MZ Miskin[‡], DJ Durian[‡]. *Machine Learning Without a Processor: Emergent Learning in a Nonlinear Analog Network* [PNAS, 2024](#)
- [8] **S Dillavou**, JM Hanlan, H Xiao, AT Chieco, S Fulco, K Turner, DJ Durian. *Bellybutton: Accessible & Customizable Deep-Learning Image Segmentation* [Nature Sci Reports, 2024](#)
- [9] **S Dillavou**, Y Bar-Sinai, MP Brenner, and SM Rubinstein. *Contact Distribution Encodes Frictional Strength* [Physical Review E, 2022](#)
- [10] **S Dillavou**, M Stern, AJ Liu, DJ Durian. *Demonstration of Decentralized, Physics-Driven Learning*, [Editor's Choice, Physical Review Applied, 2022](#)
- [11] **S Dillavou** and SM Rubinstein. *Shear Controls Frictional Aging by Erasing Memory* [Physical Review Letters, 2020](#)
- [12] **S Dillavou**, SM Rubinstein, and JM Kolinski. *The Virtual Frame Technique: Ultrafast Imaging With Any Camera* [Optics Express, 2019](#)
- [13] **S Dillavou** and SM Rubinstein. *Nonmonotonic Aging and Memory in a Frictional Interface*, [Editor's Choice, Physical Review Letters, 2018](#)

Additional

- [14] KA Murphy, **S Dillavou**, DS Bassett. *Comparing the information content of probabilistic representation spaces* [Transactions on ML Research, 2025](#)
- [15] AJ Gerra^{††}, CC Jones^{††}, **S Dillavou**, JM Hanlan, J Radzio, PE Arratia, DJ Durian. *The Equation of Motion for Taut-Line Buzzers* [Physical Review Applied, 2024](#)
- [16] T Martin, **S Dillavou**. *Calculations Without Math: "Smart instruments" & the transposition of complex shapes in the wooden boat workshop* [J of Cultural Cognitive Science, 2024](#)
- [17] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. *Training self-learning circuits for power-efficient solutions* [APL Machine Learning, 2024](#)
- [18] W Steinhardt, **S Dillavou**, M Agajanian[†], SM Rubinstein, EE Brodsky. *Seismological Stress Drops for Confined Ruptures are Invariant To Normal Stress* [Geophysical Research Letters, 2023](#)
- [19] A Srivastava ... **S Dillavou** ... Z Wu (100s of authors). *Beyond the Imitation Game: Quantifying & extrapolating the capabilities of language models* [Transactions on ML Research, 2023](#)
- [20] M Pasquet, ... AT Chieco, **S Dillavou**, JM Hanlan, DJ Durian, E Rio, A Salonen, D Langevin. *Aqueous foams in microgravity, measuring bubble sizes* [Comptes Rendus Mécanique, 2023](#)
- [21] SCL Durian[†], **S Dillavou**, K Markin[†], A Portales[†], BOT Maldonado, WTM Irvine, PE Arratia, DJ Durian. *Spreading Dynamics for Partially Wetting Droplets* [Physics of Fluids, 2022](#)
- [22] JF Wycoff[†], **S Dillavou**, M Stern, AJ Liu, DJ Durian. *Learning Without a Global Clock: Asynchronous Learning in a Physics-Driven Learning Network* [Journal of Chemical Physics, 2022](#)
- [23] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. *Physical Learning Beyond the Quasistatic Limit* [Physical Review Research, 2022](#)
- [24] S Zheng, **S Dillavou**, JM Kolinski. *Air Mediates the Impact of a Compliant Hemisphere on a Rigid Smooth Surface* [Soft Matter, 2021](#)
- [25] T Pilvelait[†], **S Dillavou**, and SM Rubinstein. *Influences of Microcontact Shape on the State of a Frictional Interface* [Physical Review Research, 2020](#)
- [26] JL Silverberg, **S Dillavou**[†], L Bonassar, and I Cohen. *Anatomic Characterization of Depth-Dependent Mechanical Properties in Neonatal Bovine Articular Cartilage* [J Orthopaedic Res, 2012](#)

Conference Workshop Proceedings

- [a] **S Dillavou**, B Beyer[†], M Stern, MZ Miskin, AJ Liu, DJ Durian. *Nonlinear Classification Without a Processor* [NeurIPS ML with New Compute Paradigms Workshop, 2023](#)
- [b] M Stern, **S Dillavou**, D Jayaraman, DJ Durian, AJ Liu. *Contrastive power-efficient physical learning in resistor networks* [NeurIPS ML with New Compute Paradigms Workshop, 2023](#)
- [c] **S Dillavou**, B Beyer[†], M Stern, MZ Miskin, AJ Liu, DJ Durian. *Circuits that train themselves: decentralized, physics-driven learning*, [Proceedings SPIE, AI & Optical Data Sciences IV, 2023](#)
- [d] M Stern, **S Dillavou**, MZ Miskin, DJ Durian, AJ Liu. *Out of Equilibrium Learning Dynamics in Physical Allosteric Resistor Networks*, [NeurIPS ML & the Physical Sciences Workshop, 2021](#)

Patents

[US Patent No 12462202 B2](#) **S Dillavou**, M Stern, MZ Miskin, AJ Liu, DJ Durian *Coupled Networks for Physics-Based Machine Learning* Nov 4, 2025

Software Packages Authored

Bellybutton – a Python deep learning package for image segmentation, designed for researchers with no coding required. Download available [here](#). Associated publication: [8]

TEACHING and MENTORING EXPERIENCE

Research Mentorship	Career Stage, University	[Coauthored Publications]	Year(s)
Jacob F Wycoff	Postgrad	[1][22]	2025-26
Alex Roseman	Undergraduate, Yale U		2024
Juan Mendez	Undergraduate, Williams		2024
Evan Stocker	Undergraduate, Pennsylvania State U		2024
Benjamin D Beyer	Undergraduate, U of Pennsylvania	[7][a][c]	2021-24
Jesse M Hanlan	Ph.D. Student, U of Pennsylvania	[4][5][8][15][20]	2020-24
Josue D Ruiz	Undergraduate, U of Pennsylvania		2022-23
Jacob F Wycoff	Undergraduate, U of Pennsylvania	[1][22]	2021-23
Courtney C Jones	Undergraduate, U Maryland	[15]	2022
Alex Gerra	Undergraduate, Moravian U	[15]	2022
Kwame Markin	Undergraduate, Swarthmore	[21]	2021
Adrian Portales	Undergraduate, U Texas Rio Grande	[21]	2021
Sylvia CL Durian	Undergraduate, U Chicago	[21]	2021
Mary Agajanian	Undergraduate, Harvard U	[18]	2019-21
Tom Pilvelait	Undergraduate, Harvard U	[25]	2018-20
Vincent Stin	Undergraduate, ESPCI, Paris		2019
Aijie Xu	Ph.D. Student, Tsinghua U		2017-18
Evgeni Shirman	Undergraduate, Hebrew U, Jerusalem		2016

Teaching

Substitute Lecturer: Analytical Mechanics, Douglas Durian - 30 Undergrads U Penn, Spring 2024

TA: Introduction to Fluid Mechanics, Shmuel Rubinstein - 60 Undergrads Harvard U, Spring 2018
 Develop new materials, in-class demos, grade assignments, supervise labs and final projects
Received Bok Center Certificate of Teaching Excellence

TA: Introduction to Soft Matter – Shmuel Rubinstein - 20 Grad Students Harvard U, Fall 2015
 Write problem sets, develop new materials, teaching section, grading

Short Courses / Workshops / Tutoring

<u>Improving Presentation & Discussion Through Improvisation</u> - 15 Grad.	Harvard U, Winter 2019
<u>Intro to Long-Form Improvisation</u> - 15 Graduate Students	Harvard U, Winter 2016
Developed and taught custom curricula for both mini-courses	
<u>Improvisational Theater Workshops</u>	Harvard, Tufts, Yale, Cornell, Deloitte, 2012-Present
Designed & taught for 5-50 participants, 6 th grade to grad, business professionals, faculty.	
<u>High school/college math/physics, SAT prep.</u>	tutored 100s of hours for 30+ students 2010-Present

Pedagogical Training

<u>Teaching and Communicating Physics</u>	Harvard U, Spring 2015
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PRESENTATIONS and PRESS

[some titles truncated for space]

Invited Talks

Analog Hardware That Trains Itself	Frontiers of Neuromorphic Computing II, Max Planck, 2026
Emergent Learning in Analog Electronic Networks	(AI) ² Research Talk Series, Princeton U, 2025
Self-Learning Networks in the Lab	Beyond von Neumann Computing Wkshp, Aspen, CO, 2025
M.L. as an Experimental Guide	PREM20: Self-Driving Science: AI & AE, San Juan, PR, 2025
Emergent Learning in Analog Electronic Networks	Physics/Math Special Seminar, NYU, 2025
The Effects of Physicality on Emergent Learning	APS March Meeting, Anaheim, CA, 2025
How to Construct a Learning 'Material' (Tutorial)	APS March Meeting, Anaheim, CA, 2025
Emergent Physical Learning	Applied Physics Special Seminar, Cornell U, 2024
Emergent Physical Learning	Tufts Condensed Matter Seminar, 2024
Emergent Machine Learning	Inaugural klogW Future Series Seminar, Virtual, 2024
The Metamaterial that Trains Itself	SIAM: Mathematical Aspects of Mat Sci, Pittsburgh, PA, 2024
Emergent Learning in Electronic Networks	Physics Dept Special Seminar, U Chicago, 2024
Supervised Learning as an Emergent Property	Argonne Nat'l Lab Applied AI Series, Virtual, 2023
Evolution of a Learning Material	9 th IDMRCS, Chiba, Japan, 2023
Decentralized, Physics-Driven Learning	SPIE Photonics West, San Francisco, CA, 2023
A Physics-Driven Learning Network	Alternative Computing Group Seminar, NIST, 2023
Demonstration of Decentralized, Physics-Driven Learning	Phys Rev Journal Club, Virtual, 2022
Hijacking Physics to Learn for Us	Weekly Seminar, Google Brain, 2022
Using Physics to Learn without a Processor	APS March Meeting, Chicago, IL, 2022
Decentralized Physics-Driven Learning	Physics Seminar, Bucknell U, 2021
Hidden Dynamics of Static Friction	Applied Math Seminar, NYU, 2020
Hidden Dynamics of Static Friction	Soft Matter Coffee Hour, Princeton U, 2020
Hidden Dynamics of Static Friction	Soft Matter Theory Group (U Pennsylvania), Virtual, 2020
Static Friction: Aging and Memory	Geomechanics Seminar, Pennsylvania State U, 2018
Memory in Solid-Solid Interfaces	Mechanical Engineering Seminar, EPFL, 2018

Selected Press**On Physical & Emergent Learning**

Machines That Learn Like Physicists Think	APS Research Highlights, 2026
How a simple circuit could offer an alternative to energy-intensive GPUs	MIT Tech Review, 2024
A first, physical system to learn nonlinear tasks without a traditional processor	Penn Today, 2024
Training neural networks using physical equations of motion	PNAS Physics Commentary 2024
How to make the universe think for us	Quanta Magazine, 2022

Simple electrical circuit learns on its own – with no help from a computer Science News, 2022
 Programming matter to a computer's job American Physical Society News, 2021

On the Virtual Frame Technique

Imaging technique lets ordinary cameras capture high-speed images of crack formation Phys.org 2019
 How to mod a smartphone camera so it shoots a million frames per second MIT Tech Review, 2018

On Memory in Frictional Interfaces

Friction Remembers Its Origins American Physical Society Physics Focus, 2018
 Friction Remembers Its Past Physics Today, 2018

Contributed Talks

Local Learning Rules in Analog Hardware Computing with Physical Systems, 2026
 Studying Granular Clogging with ML as an Experimental Guide NE Granular Mat, Holy Cross, 2024
 Emergent Learning Via Sequential Error Mode Reduction APS March Mtg, Minneapolis, MN, 2024
1st Prize Statistical & Nonlinear Physics Postdoctoral Speaker Award
 Transistor-Based Self-Learning Networks APS March Meeting, Las Vegas, NV, 2023
 A Physics-Driven Self-Learning Transistor Network Coherent Network Comp., Stanford U, 2022
 Contact Distribution Encodes Frictional Strength APS March Meeting, Chicago, IL, 2022
 Decentralized Physics-Driven Learning Rising Stars in Soft & Biological Matter, U Chicago, 2021
 Building a Physical Learning Network APS March Meeting, Virtual, 2021
 Memory in Solid-Solid Interfaces APS March Meeting, Boston, MA, 2019
 Hidden Dynamics of Static Contact and Static Friction Dynamics Days, Northwestern U, 2019
 Extreme Mechanics of Elastomer Impact NORA & BASF Collab. Days, U Mass Amherst, 2019
 Two Solids Make a Glass: Memory in Solid-Solid Interfaces APS March Mtg, Los Angeles, CA, 2018
 Elastomer Wear: The NBA's Shoe Problem NORA & BASF Collab. Days, U Mass Amherst, 2017
 Memory in the Frictional Interface Physics Dept Mini-Symposium, Weizmann Inst, 2017

Posters/Rapid Talks

Nonlinear Classification Without a Processor Ctr for Soft and Living Matter Kickoff, UPenn 2024
 Self-Learning Electronic Networks Mid Atlantic Soft Matter Workshop, Georgetown U 2024
 Nonlinear Classification Without a Processor Computing with Physical Systems, Aspen, CO 2024
 A Physics-Driven Self-Learning Transistor Network APS March Meeting, Las Vegas, NV 2023
1st Prize in the APS March Meeting Postdoctoral Poster Competition
 Physical Learning Machines Cracking the Glass Problem Simons Mtg, New York, NY 2022
 Building a Physical Learning Network Northeast Complex Fluids Workshop, Virtual 2021
 Tabletop Nucleation Southern California Earthquake Center Annual Mtg, Palm Springs, CA 2019
 The Hidden Dynamics of Static Friction Gordon Conf: Soft Matter Phys., New London, NH 2019
 The Virtual Frame Technique 77th New England Complex Fluids, Harvard U 2018
 Memory in the Frictional Interface 73rd New England Complex Fluids, Harvard U 2018
 Memory in the Frictional Interface Jay (Fineberg) Fest, Hebrew U in Jerusalem, 2017
 Beyond Rate and State: Frictional Memory Inst. for Study of the Continents Conf, Cornell U, 2017
 Wear in Basketball Shoes NORA & BASF Challenges, U Mass Amherst, 2017
 Visualizing Frictional Interfaces 69th New England Complex Fluids, Harvard U 2016
 Visualizing Frictional Interfaces 67th New England Complex Fluids, MIT 2016
 Loading History of Frictional Interfaces Phys + Mech of Soft Complex Mat, Cargese, France, 2016
 Loading History of Frictional Interfaces Gordon Conference: Tribology, Lewiston, ME, 2016
 Visualizing Growth of a Multi-contact Interface 65th NE Complex Fluids, Harvard U 2015
 Aging of Multi-Contact Interfaces Soft Matter: Friction, Rheology, Tribology, U Florida 2015

PROFESSIONAL SERVICE

Journal Referee

Science, Nature [Communications, Electronics], Soft Matter, Physical Review [B, E, Appl., Letters],
US Geological Survey Internal, J of Geophysical Res - Solid Earth, NeurIPS Workshops

Outreach

Design and teach U Penn Data Driven Discovery Initiative ML Workshop	2023, 2024, 2025
Philly Materials Day (K-12), design, construct and demo trainable elastic material	2024
Design and teach U Penn REU Machine Learning (ML) Workshop	2022, 2023, 2024
DEEPenn STEM (see below), volunteer, mentor, presenter	2023
Science Café speaker, Wilmington, Delaware. <i>"Friction: The surprising unsolved science behind earthquakes and tire treads"</i>	2023
Planning committee, volunteer, presenter for the first annual DEEPenn STEM: weekend-long STEM PhD prep/info workshop for ~45 URM college students from around the country	2022
2 nd Place, MRSEC National Science Slam: Learning Networks on the Radio	2022
<i>Science in the News</i> Writer, Harvard U	2016-17
<i>Splash at Yale</i> Instructor, grades 7-9 and 10-12, Yale U	2016, 2017

Professional Membership

American Physical Society	2016-Present
APS March Meeting session organizer, chair, and sorter	2023, 2024

Misc: Part of a collaboration developing a [3D Printer-as-Ventilator](#) during COVID-19 outbreak 2020